

Physical Completeness vs. Statistical Identifiability

A reading note on *Finite Observation and Broader Physical Processes* and *When Does a Random-Scale Flight Path Look Lognormal?*

The two papers ask different questions about what a good-looking fit means. The first asks whether a finite fit establishes that the visible physical process is the whole system; the second asks whether the distribution of observed flight trajectories is compatible with one proposed statistical population.

The distinction

- **Finite Observation: a good finite fit does not establish physical completeness.** The example compares one visible harmonic oscillator with the same oscillator weakly coupled to a second one. During any fixed experiment, the coupling can be small enough that the two models produce nearly indistinguishable records, even though energy exchange with the second oscillator creates a large difference later:

$$\text{TV}(P_\delta^O, P_0^O) \leq \varepsilon, \quad |q_\delta(T_*) - q_0(T_*)| = A.$$

Here P_δ^O and P_0^O are the probability laws of all recorded measurements under the coupled and isolated models, using the same observation protocol O . The coupling-induced frequency split is $\delta > 0$, TV is total-variation distance, and ε is the chosen distinguishability tolerance. The corresponding visible trajectories are q_δ and q_0 ; at the later time T_* they differ by the full amplitude A . The result does *not* infer that such a larger process exists in any particular observed system. It says that finite predictive success alone cannot rule one out or certify that a physical model is complete.

The paper then states separately what must be assumed to say that further relevant scales actually exist. If $N(R, S)$ counts processes with characteristic scale $L \in (R, S]$, and their Poisson intensity is $d\eta(L) dL/L$, then

$$\Pr\{N(R, S) \geq 1\} = 1 - \exp\left[-d \int_R^S \frac{\eta(L)}{L} dL\right], \quad \int_R^\infty \frac{\eta(L)}{L} dL = \infty$$

is the condition for unbounded scales almost surely. Here $R < S$ are scale cutoffs, d converts scale to volume, and $\eta(L)$ is the assumed intensity per logarithmic-volume increment. This is a conditional population statement, not something established by the finite record.

- **The aerostat paper: skewness and kurtosis reveal when one statistical population is inadequate.** The proposed model factors each track's RMS deviation from its endpoint geodesic as

$$D = AQ, \quad \log D = \log A + \log Q,$$

where $D > 0$ is observed RMS deviation, $A > 0$ is overall route scale, and $Q > 0$ is the RMS size of the dimensionless endpoint-pinned path shape. If $\log A$ is Gaussian and independent of $\log Q$, variation in A should wash out the shape's non-Gaussian higher moments. With

$$\rho = \frac{\tau_Q^2}{\sigma_A^2 + \tau_Q^2}, \quad \sigma_A^2 = \text{Var}(\log A), \quad \tau_Q^2 = \text{Var}(\log Q),$$

the model predicts

$$\text{Skew}(\log D) = \text{Skew}(\log Q)\rho^{3/2}, \quad \text{ExKurt}(\log D) = \text{ExKurt}(\log Q)\rho^2.$$

Here Var is variance, Skew is standardized skewness, and ExKurt is excess kurtosis; ρ is the fraction of total log-variance contributed by shape. After matching the observed variance, the single-population Brownian-bridge model predicts skewness about 0.003 and excess kurtosis about -0.002. The data instead give -2.40 and 8.11. This sharp mismatch statistically rejects the single-population explanation and motivates a near-geodesic boundary population or mixture. It detects missing statistical heterogeneity; it does not identify its physical cause or fully determine the mixture components.

Only after that population-level rejection does the paper ask when one flight can be assigned to the boundary component. If $C = 0$ denotes that component, $\pi_0 = \Pr(C = 0)$ is its prior proportion, and $R_n^2 = n^{-1} \sum_{j=1}^n \eta(t_j)^2$ is prefix RMS deviation after n observations at times t_j , with cross-track deviations $\eta(t_j)$, then for a small-deviation threshold τ ,

$$\Pr(C = 0 \mid R_n \leq \tau) \geq \frac{\pi_0(1 - \beta_n)}{\pi_0(1 - \beta_n) + (1 - \pi_0)e^{-nI(\tau)}},$$

$$n \gtrsim \frac{\log((1 - \pi_0)/\pi_0) + \log((1 - \alpha)/\alpha)}{I(\tau)}.$$

Here β_n bounds boundary misses, $I(\tau) > 0$ is the ordinary population's exponential small-deviation rate, and α is the tolerated posterior error; $\gtrsim$ gives the waiting-time scale up to the β_n correction. This follow-on result gives the time to diagnose an individual boundary track—about 41 minutes in the dataset—rather than the original evidence that a mixture is needed.

Finite Observation: Does a good finite fit prove that no larger coupled physical process matters? No.

Aerostat: Do the observed skewness and kurtosis agree with the proposed single-population trajectory model? No—their mismatch makes additional statistical structure necessary.

The papers are therefore complementary: the first warns that even an excellent finite fit cannot certify physical closure; the second shows how observable higher moments can nevertheless falsify a specific statistical explanation and reveal population heterogeneity.

Companion papers: *Finite Observation and Broader Physical Processes*; *When Does a Random-Scale Flight Path Look Lognormal?* (J. R. Landers, 2026).